

Flight Price Prediction

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[GitHub Repository](#)

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Data Workflow

- Amadeus API
 - Limit – 10,000 monthly API calls per month per API key
- Scheduling via GitHub Workflows
 - Automates runs for data collection over time
- Our .py script runs 2x per day, collection ~ 24,000 flight observations per day
- The script saves each run as a new .csv file in Dropbox, which we can access through a Dropbox API call
- We have been running this pipeline since 11/10
 - Some issues cause us to miss some days

Final Data

Data Collection Metrics:

- Total observations: 679,621
- Unique flights tracked: 35,119
- Flights with price changes: 31,612
- Average price change: \$190.97
- Collection period: 2025-11-10 to 2025-12-17

- Our dataset for modeling and EDA has:
 - ~700,000 observations of >35,000 unique flights
 - Average flight's price changed by almost \$200 over the collection period
- Amadeus features:
 - Price in Euros (transformed to USD)
 - Flight length (via departure and arrival times)
 - Seats remaining
 - Days until departure
 - Aircrafts being used
 - Number of stops

Aircraft Dashboard

- Features

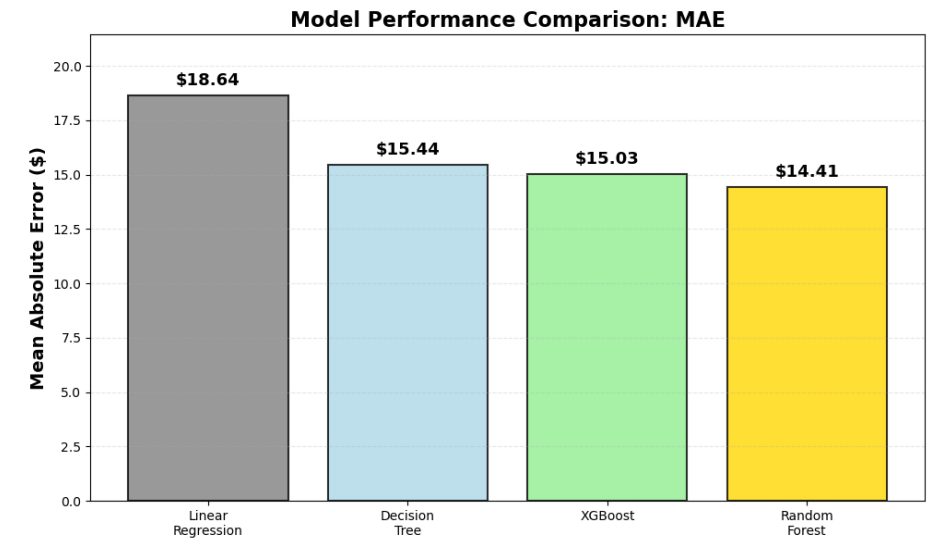
- Built using Dash framework
- Analyzes all flight observations
- Filter systems
 - Basic: Airline, Route, Aircraft Details
 - Added: Origin, Destination, Aircraft Brand
 - Flight Details: Stops, Cabin, Red-eye (NEW)
 - Advanced: Price, Days until Departure, Departure time, Flight Duration, Bookable seats
- Visual Analytics: Interactive pie chart

ML Models - Introduction

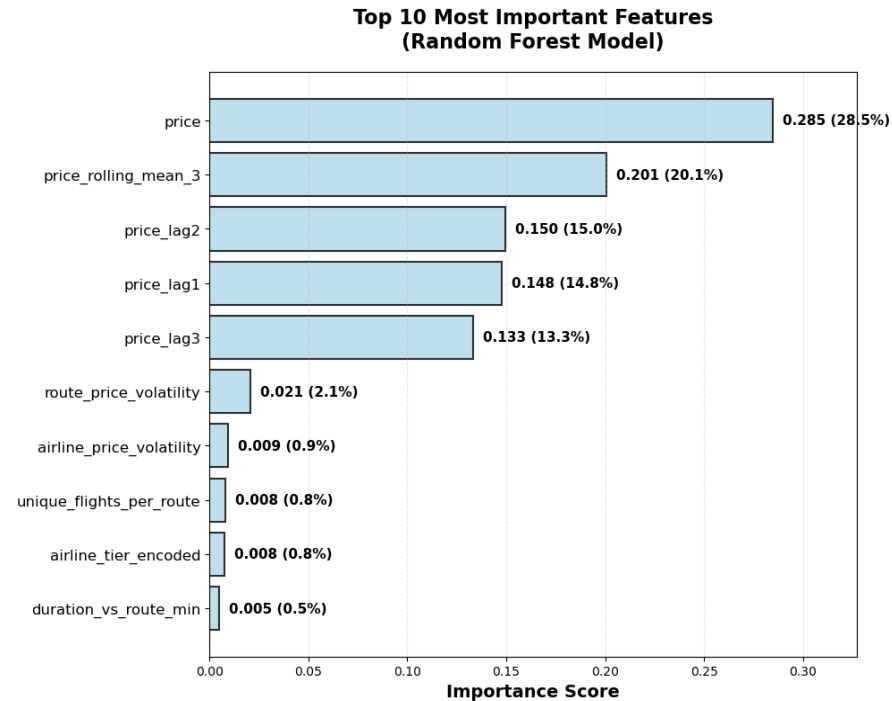
- Research Question: **Using machine learning, can we reliably predict what a cross-country flight's price will be *tomorrow*?**
- Feature Engineering:
 - Modeling features were derived from Amadeus features to improve models
 - Some features created:
 - Lagged prices (price yesterday, 2 days ago, etc.)
 - Rolling window prices (average price over the past 3 days)
 - Temporal features (day of week, days before departure, etc.)
 - Other features (is_4_seats_remaining, route/airline volatility, etc.)
 - Why these features?
 - Capture price trends over time, booking window trends, and competition indicators
- Modeling Approach:
 - 80/20 temporal split
 - Hyperparameter tuning (GridSearchCV & RandomizedSearchCV)
 - 3-fold cross-validation
- Models Tested:
 - Linear Regression, Decision Tree, Random Forest, XGBoost

ML Models - Performance

- Model Evaluation Statistic:
 - Mean Absolute Error (MAE)
- Linear Regression:
 - Average prediction is \$18.64 away from actual price
- Tree Models show improved performance
 - Indicates non-linear relationships between features and price
- Best Model: Random Forest
 - MAE of 14.41 (average prediction within \$15 from actual price)
 - R^2 of 0.991



ML Models – Feature Importance



- Most important features:
 - Price history captures ~91.7% of predictive power
 - Current price, lagged price, rolling averages
 - Model learns price trajectory, not just the current price
 - Volatility by route and airline (~3% importance score)
 - Route-based features (duration and competition)

Conclusions & Implications

- Prediction Capability:
 - Random Forest Model predicts tomorrow's price within \$14.41 on average
 - Highly accurate, allowing for confident booking decisions
- Social/Economic Implications:
 - We can help people find the best day to book a cross-country flight
 - Consumers can save money on expensive flights
- Ethical & Legal Implications:
 - Airline pricing strategies are private
 - Highly predictive models could uncover these secrets
 - Multiple API keys used to get around costs and limits
 - Legal for academic project, not for selling an algorithm or insights

Limitations & Continuation

- Limited by API limitations
 - Only 12,000 observations per run
 - Only select airlines
 - Only select airports
- Temporal limitations
 - Data collection period: 11/10 - today (missing 1 week-long period for API issues)
 - Time period is Thanksgiving – Christmas time (possibly unusual patterns)
- Continuation:
 - Longer forecasting (only 3-day lags created)
 - Adding external factors (weather, city events, etc.)
 - Larger time frame
 - More routes, cabin classes, airlines, etc.